

HW1 Solution

1. For the TLC trial considered, express the mean response in each group on each occasion in terms of the regression coefficients $\beta_1, \dots, \beta_8$.
- 2.

Treatment	Week 0	Week 1	Week 4	Week 6
Succimer	$\beta_1 + \beta_2$	$\beta_1 + \beta_2 + \beta_3 + \beta_6$	$\beta_1 + \beta_2 + \beta_4 + \beta_7$	$\beta_1 + \beta_2 + \beta_5 + \beta_8$
Placebo	β_1	$\beta_1 + \beta_3$	$\beta_1 + \beta_4$	$\beta_1 + \beta_5$

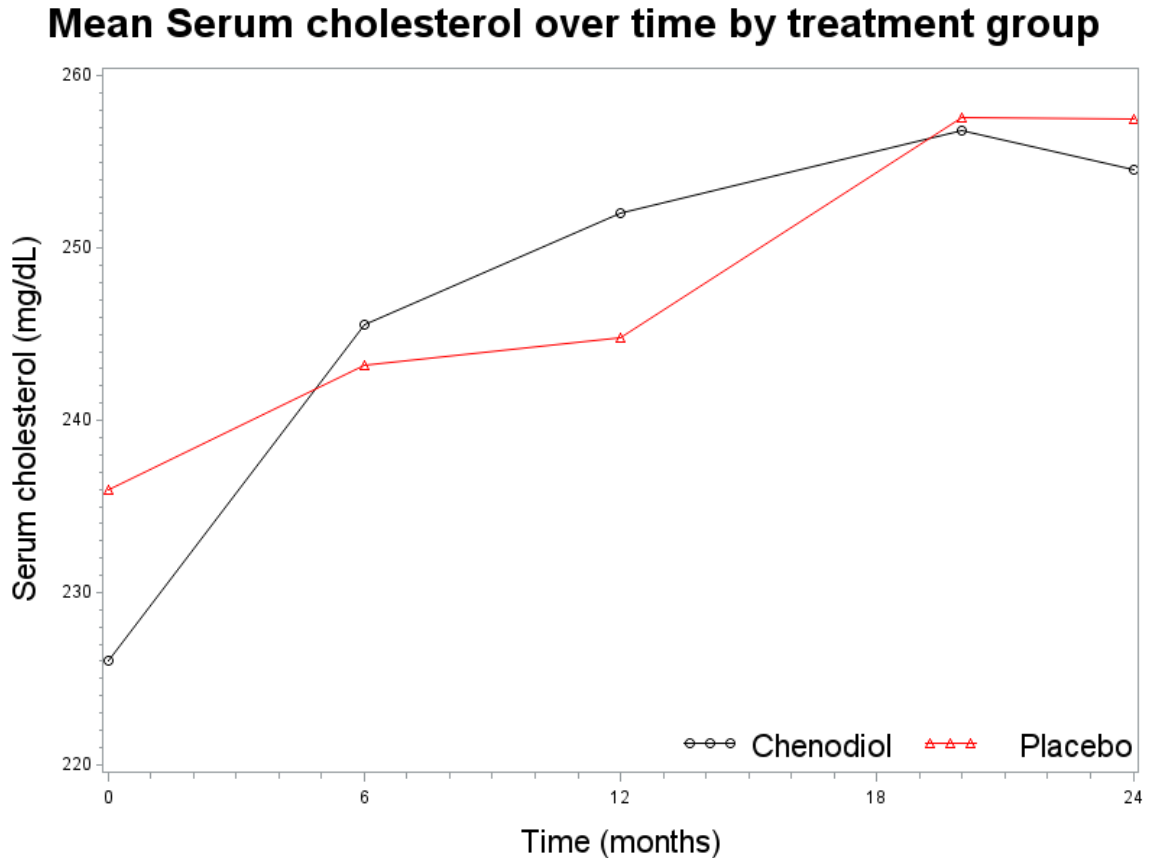
5.1.1 Read the data from the external file and keep it in a “multivariate” or “wide” format.

5.1.2. Calculate the sample means, standard deviations, and variances of the serum cholesterol levels at each occasion for each treatment group.

Table 1. Summary of serum cholesterol measurements (mg/dL) at baseline and over follow-up by treatment group.

	<i>High dose chenodiol</i>				<i>Placebo</i>			
	N	Mean	Std Dev	Variance	N	Mean	Std Dev	Variance
<i>Baseline</i>	62	226.0	39.7	1573.3	41	235.9	55.9	3122.0
<i>Month 6</i>	62	245.5	39.5	1556.5	41	243.2	49.2	2424.6
<i>Month 12</i>	55	252.0	38.3	1469.1	38	244.8	46.1	2126.2
<i>Month 20</i>	44	256.8	34.5	1189.5	35	257.6	51.1	2615.5
<i>Month 24</i>	38	254.6	50.0	2496.2	31	257.5	49.4	2439.2

5.1.3. On a single graph, construct a time plot that displays the mean serum cholesterol versus time (in months) for the two treatment group. Describe the general characteristics of the time trends for the two groups.



Overall, mean serum cholesterol tends to increase over time in both groups. The high-dose chenodiol group starts with a lower value at baseline but surpasses the placebo group over months 6 and 12. The two groups roughly converge at month 20. Then, the chenodiol group decreases slightly at month 24 while the placebo group remains unchanged.

5.1.4. Next read the data from the external file and put the data in a “univariate” or “long” format with five records per subject.

5.1.5. Assuming an unstructured covariance matrix, conduct an analysis of response profiles. Determine whether the patterns of change over time differ in the two treatment groups.

To conduct an analysis of response profiles, we build the following model:

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 \\ + \beta_7 X_2 X_3 + \beta_8 X_2 X_4 + \beta_9 X_2 X_5 + \beta_{10} X_2 X_6 + e$$

where the response Y is serum cholesterol (mg/dL), $X_2 = 1, 0$ indicates chenodiol and placebo groups, respectively, X_3--X_6 are indicator variables for months 6, 12, 20, and 24, respectively, and e is the random error term. To determine whether the patterns of change differ between groups, we test the null hypothesis on the interaction terms:

$$H_0: \beta_7 = \beta_8 = \beta_9 = \beta_{10} = 0.$$

The chi-square test statistic is 7.86. With reference to the chi-square distribution with 4 degrees of freedom, the p value is 0.0968. Thus, we conclude that there is no significant difference in the pattern of change at the 0.05 level.

5.1.6. Display the estimated 5x5 covariance and correlation matrices for the five repeated measurements of serum cholesterol.

Table 2a. Estimated covariance matrix for response at different occasions.

COVARIANCE	MONTH 0	MONTH 6	MONTH 12	MONTH 20	MONTH 24
MONTH 0	2186.6	1570.1	1407.5	1444.5	1326.2
MONTH 6	1570.1	1900.3	1390.5	1455.0	1406.0
MONTH 12	1407.5	1390.5	1692.7	1250.1	1347.7
MONTH 20	1444.5	1455.0	1250.1	1756.3	1264.2
MONTH 24	1326.2	1406.0	1347.7	1264.2	2321.3

Table 2b. Estimated correlation matrix for response at different occasions.

CORRELATION	MONTH 0	MONTH 6	MONTH 12	MONTH 20	MONTH 24
MONTH 0	1.000	0.770	0.732	0.737	0.589
MONTH 6	0.770	1.000	0.775	0.796	0.670
MONTH 12	0.732	0.775	1.000	0.725	0.680
MONTH 20	0.737	0.796	0.725	1.000	0.626
MONTH 24	0.589	0.670	0.680	0.626	1.000

5.1.7. With baseline (month 0) and the placebo group (group 2) as the reference, write out the regression model for mean serum cholesterol that corresponds to the analysis of response profiles in Problem 5.1.5.

See answer to Problem 5.1.5.

5.1.8. Let L denote a matrix of known weights and β the vector of linear regression parameters from the model assumed in Problem 5.1.7. The null hypothesis that the patterns of change over time do not differ in the two treatment groups can be expressed as $H_0: L\beta = 0$. Describe an appropriate weight matrix L for this null hypothesis.

Write $\beta = (\beta_1, \dots, \beta_{10})'$. The null hypothesis is

$$H_0: \begin{pmatrix} \beta_7 \\ \beta_8 \\ \beta_9 \\ \beta_{10} \end{pmatrix} = 0.$$

This is equivalent to $L\beta = 0$, where

$$L = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

5.1.9. Show how the estimated regression coefficients from an analysis of response profiles can be used to construct the time-specific means in the two groups. Compare these estimated means with the sample means obtained in Problem 5.1.2.

	Baseline	Month 6	Month 12	Month 20	Month 24
Placebo	β_1	$\beta_1 + \beta_3$	$\beta_1 + \beta_4$	$\beta_1 + \beta_5$	$\beta_1 + \beta_6$
Chenodiol	$\beta_1 + \beta_2$	$\beta_1 + \beta_2 + \beta_3 + \beta_7$	$\beta_1 + \beta_2 + \beta_4 + \beta_8$	$\beta_1 + \beta_2 + \beta_5 + \beta_9$	$\beta_1 + \beta_2 + \beta_6 + \beta_{10}$

The time-specific means computed by regression coefficients are very similar to the raw means obtained in Problem 5.1.2. This is not surprising because we used a saturated mean model for the analysis of response profiles. The minor differences between the model-based and raw means are only due to missing data.

5.1.10. With baseline (month 0) and the placebo group (group 2) as the reference group, provide an interpretation for each of the estimated regression coefficients in terms of the effect of the treatments on the patterns of change in mean serum cholesterol.

PARAMETER	ESTIMATE	INTERPRETATION
β_1	235.9	Mean baseline response in placebo
β_2	-9.9	Difference in mean baseline response between Chenodiol and placebo
β_3	7.2	Change from baseline to month 6 in placebo
β_4	8.8	Change from baseline to month 12 in placebo
β_5	23.1	Change from baseline to month 20 in placebo
β_6	21.1	Change from baseline to month 24 in placebo
β_7	12.3	Difference in change from baseline to month 6 between Chenodiol and placebo
β_8	16.4	Difference in change from baseline to month 12 between Chenodiol and placebo
β_9	5.0	Difference in change from baseline to month 20 between Chenodiol and placebo
β_{10}	6.9	Difference in change from baseline to month 24 between Chenodiol and placebo